Week 5 Progress

Explainable Multi-Label Content-Based Image Retrieval for Aerial Images using Classical Computer Vision

Group 16

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1. Feature Normalization and Similarity Measures

- The existing Engine v2 was tested with Z-score normalization to check whether normalizing the features would improve retrieval.
- Normalization reduced Class Precision@10 from 0.4336 to 0.4030 and the selection score from 0.3970 to 0.3813.
- Retuning the weights after normalization recovered some performance, but the result was still below the original Engine v2.
- Therefore, Z-score normalization was dropped.
- Different similarity measures were tested for the colour and texture features.
- Hellinger similarity gave better results for HSV, LBP radius 1 and LBP radius 2.
- Cosine similarity was retained for HOG.

2. Feature Weight Tuning

- The weights of HSV, LBP radius 1, LBP radius 2 and HOG were retuned using cross-validation.
- Five-fold leave-one-fold-out (LOFO) cross-validation was used to compare the feature combinations.
- The same weight combination was selected across all five folds.
- The final feature weights were:

Feature	Weight
HSV	0.20
LBP radius 1	0.40
LBP radius 2	0.20
HOG	0.20

- This configuration was selected as the V3 baseline.
- V3 was frozen before starting the label-context experiments.

3. V3 Results

Metric	V2	V3	Change
Class P@10	0.4336	0.5069	+0.0733
ML P@10	0.4080	0.4556	+0.0475
ML mAP	0.2411	0.2738	+0.0327
nDCG@10	0.6901	0.7286	+0.0385
Selection score	0.3970	0.4439	+0.0469

- V3 performed better than V2 across all five validation folds.
- Class Precision@10 increased from 0.4336 to 0.5069.
- Multi-label mAP increased from 0.2411 to 0.2738.
- The selection score increased from 0.3970 to 0.4439.
- V3 was retained as the main visual-only retrieval system.

4. Label-Context Experiments

- After freezing V3, label context was tested to check whether predicted labels could improve image retrieval.
- Labels were predicted for a query image using its neighbouring images.
- The predicted-label similarity was combined with the visual similarity using the parameter λ.
- Different values of λ and different neighbourhood sizes ($k = 3, 5, 7$) were tested.
- Hard voting, soft voting and different thresholds were also evaluated.
- Label-only and oracle experiments were included for comparison.
- The experiments were evaluated using the same five-fold LOFO cross-validation approach.

5. Label-Context Results

System	Class P@10	ML P@10	ML mAP	Selection score
Frozen V3	0.5069	0.4556	0.2738	0.4439
Visual-optimal	0.5610	0.5695	0.4103	0.5364
Balanced-optimal	0.5494	0.5914	0.5012	0.5647
Selection-optimal	0.5267	0.5963	0.5971	0.5771
Label-only	0.3757	0.5921	0.5998	0.5359

- Label context improved multi-label retrieval performance compared with the frozen V3 baseline.
- The selection-optimal configuration achieved the highest selection score of 0.5771.
- The balanced-optimal configuration achieved a selection score of 0.5647 while maintaining a higher Class Precision@10 than the selection-optimal configuration.

- The visual-optimal configuration achieved the highest Class Precision@10 among the tested label-context configurations.
- The label-only experiment achieved high multi-label mAP but lower Class Precision@10, showing that label similarity alone does not preserve visual scene similarity as well.

6. Challenges Faced

- Z-score normalization reduced retrieval performance, even after retuning the weights.
- Choosing the right similarity measure was important for improving colour and texture retrieval.
- Label context improved multi-label retrieval, but too much dependence on predicted labels could reduce visual scene similarity.
- Different retrieval objectives favoured different values of λ.
- Oracle results were useful for comparison but were not suitable as a deployable system because they used ground-truth labels.

7. Week 5 Conclusion

Week 5 focused on improving the visual retrieval system and testing whether label information could further improve its performance. Z-score normalization was dropped because it reduced the results. Hellinger similarity was selected for HSV and both LBP features, and the feature weights were retuned using cross-validation. This produced a stronger V3 baseline, which was then frozen. Label-context experiments showed that predicted labels can improve multi-label retrieval, but relying too much on labels can hurt visual scene retrieval. Therefore, V3 remains the main label-free system, while label-aware retrieval is treated as a separate experiment.

Week 6 Progress Plan

Objective

The main objective for Week 6 is to improve the frozen V3 visual retrieval system using neighbourhood information without using any image labels.

1. Query Expansion

- Test query expansion using the top retrieved images to refine the original query representation.
- Check whether the refined query retrieves more relevant aerial images.
- Tune the query expansion parameters using only the training folds.

2. K-Reciprocal Re-Ranking

- Test k-reciprocal re-ranking to determine whether images that are mutual neighbours can improve the retrieval ranking.

- Use neighbourhood relationships between images to refine the initial ranking.
- Tune the re-ranking parameters using only the training folds.

3. Evaluation and Comparison

- Evaluate both methods using the same five-fold LOFO cross-validation setup.
- Compare every experiment directly against the frozen V3 baseline.
- Use Class P@10, ML P@10, ML mAP, nDCG@10 and the selection score.
- Record per-fold results and improvements or decreases (deltas) relative to V3.
- Check whether neighbourhood information provides consistent improvements or simply changes the ranking.

Expected Outcome of Week 6

By the end of Week 6, query expansion and k-reciprocal re-ranking will have been evaluated against the frozen V3 baseline. Based on the cross-validation results, we will decide whether to adopt query expansion, k-reciprocal re-ranking, both methods, or neither for the next version of the retrieval system.

Methodological Note: All parameter tuning will be performed using the training folds, and performance will be evaluated on the corresponding held-out fold. No image labels will be used in the Week 6 neighbourhood-based experiments.